

Radiation Damage of Reactor Pressure Vessel Steels Studied by Positron Annihilation Spectroscopy-A Review

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Can Hydrogen Ion Implantation Stand In for Neutron Damage?

Roadmap

Motivation

Shortcut

Measurement

Evidence

Takeaway

1

Motivation

Why Defects
Matter

2

Shortcut

H-Ion
Implantation

3

Measurement

PLEPS
Depth Scan

4

Evidence

DPA Vs H
Vs Lifetime

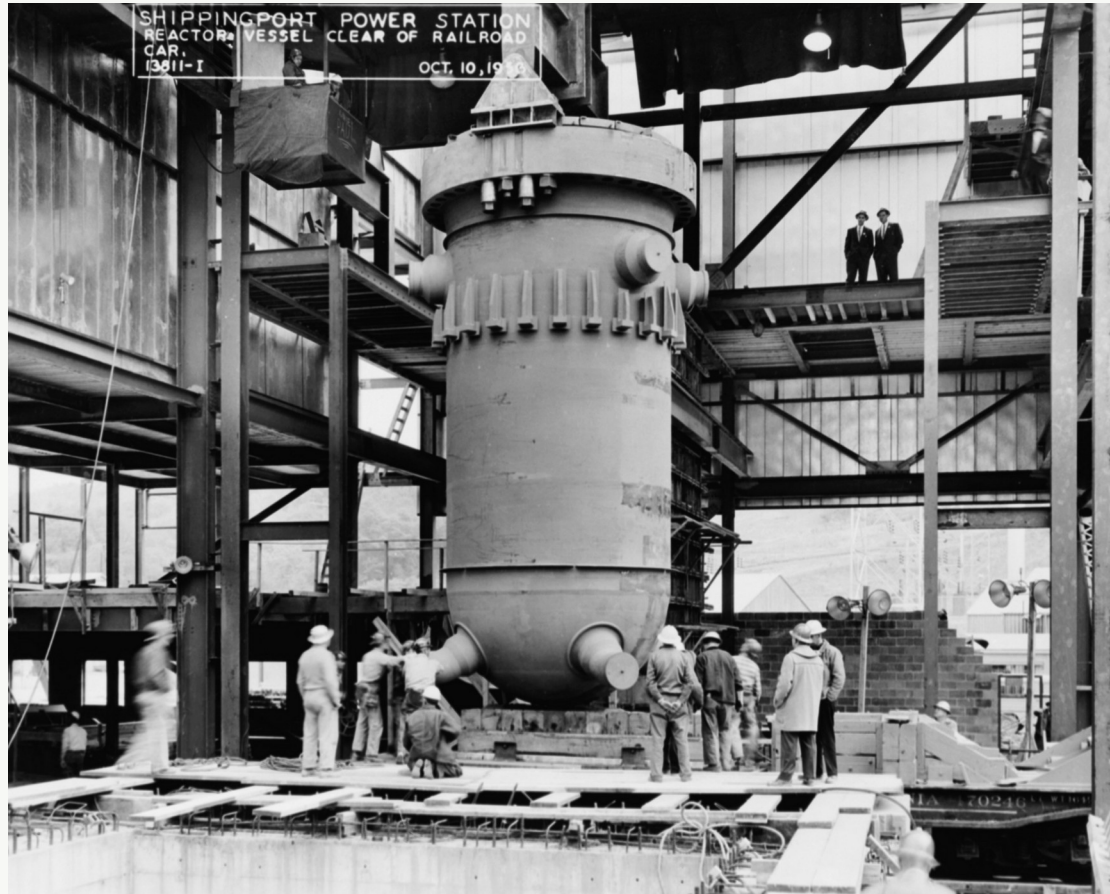
5

Takeaway

Useful, Not
A Clean Copy

Does Matching DPA Mean Matching Defect Physics?

Reactor Pressure Vessel



Invisible Defects Matter

Motivation

Shortcut

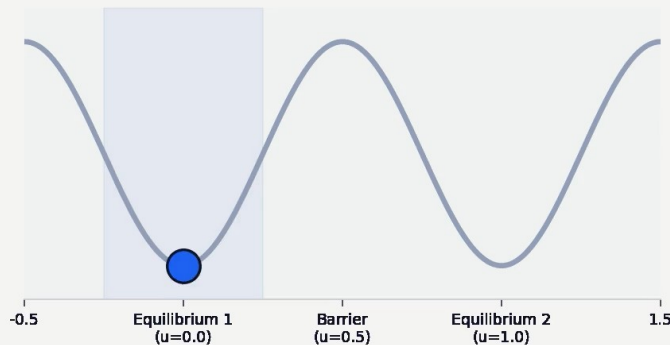
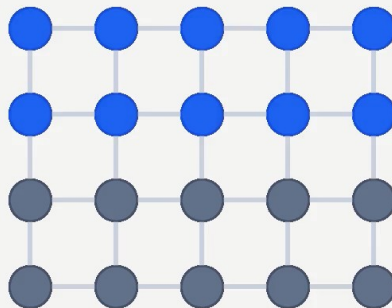
Measurement

Evidence

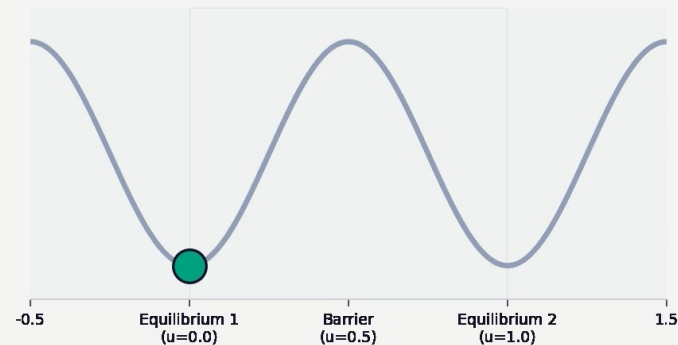
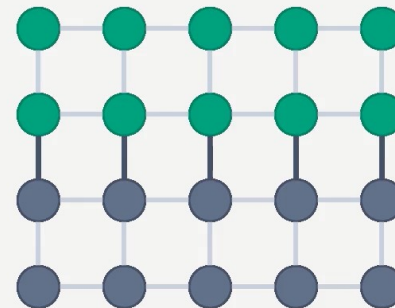
Takeaway

- Damage starts before the crack
- Steel survives stress by deforming plastically
- If it cannot yield, stress has nowhere gentle to go.

ELASTIC DEFORMATION (Reversible)

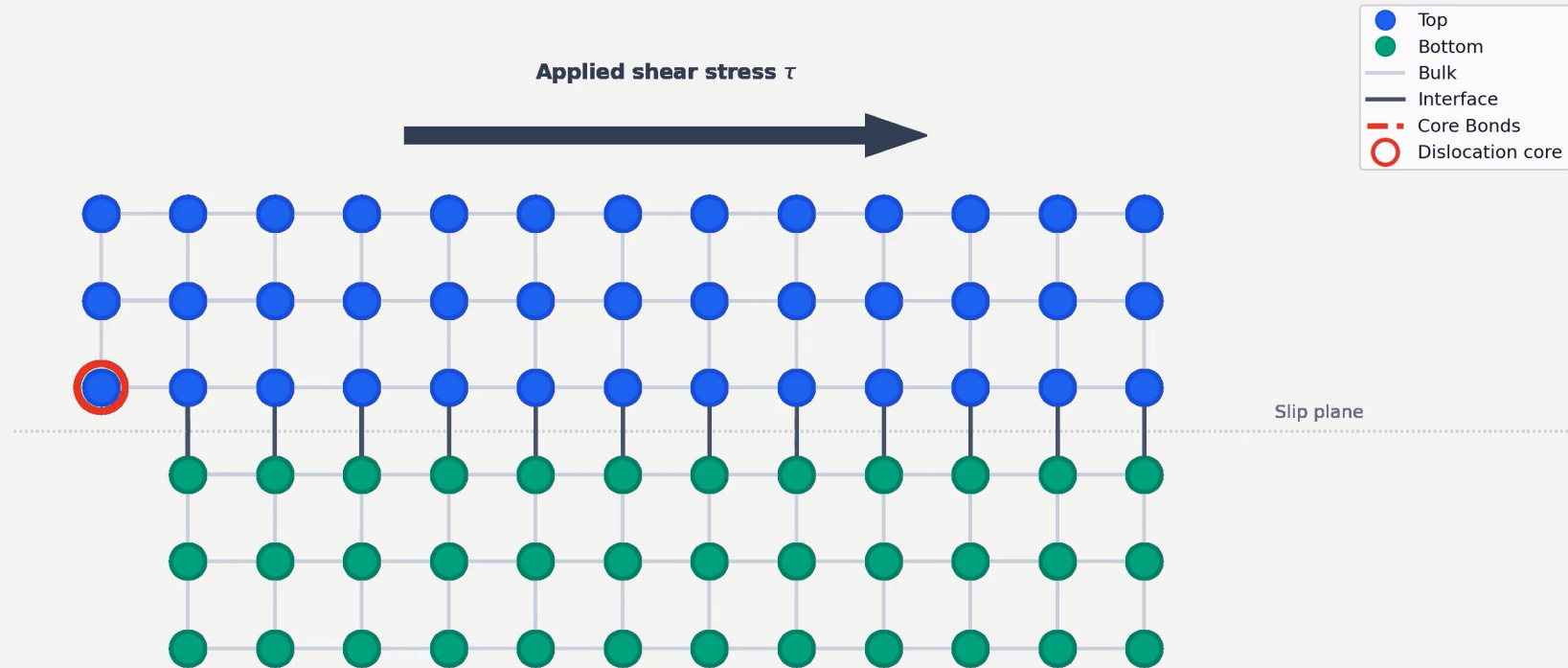


PLASTIC DEFORMATION (Irreversible)

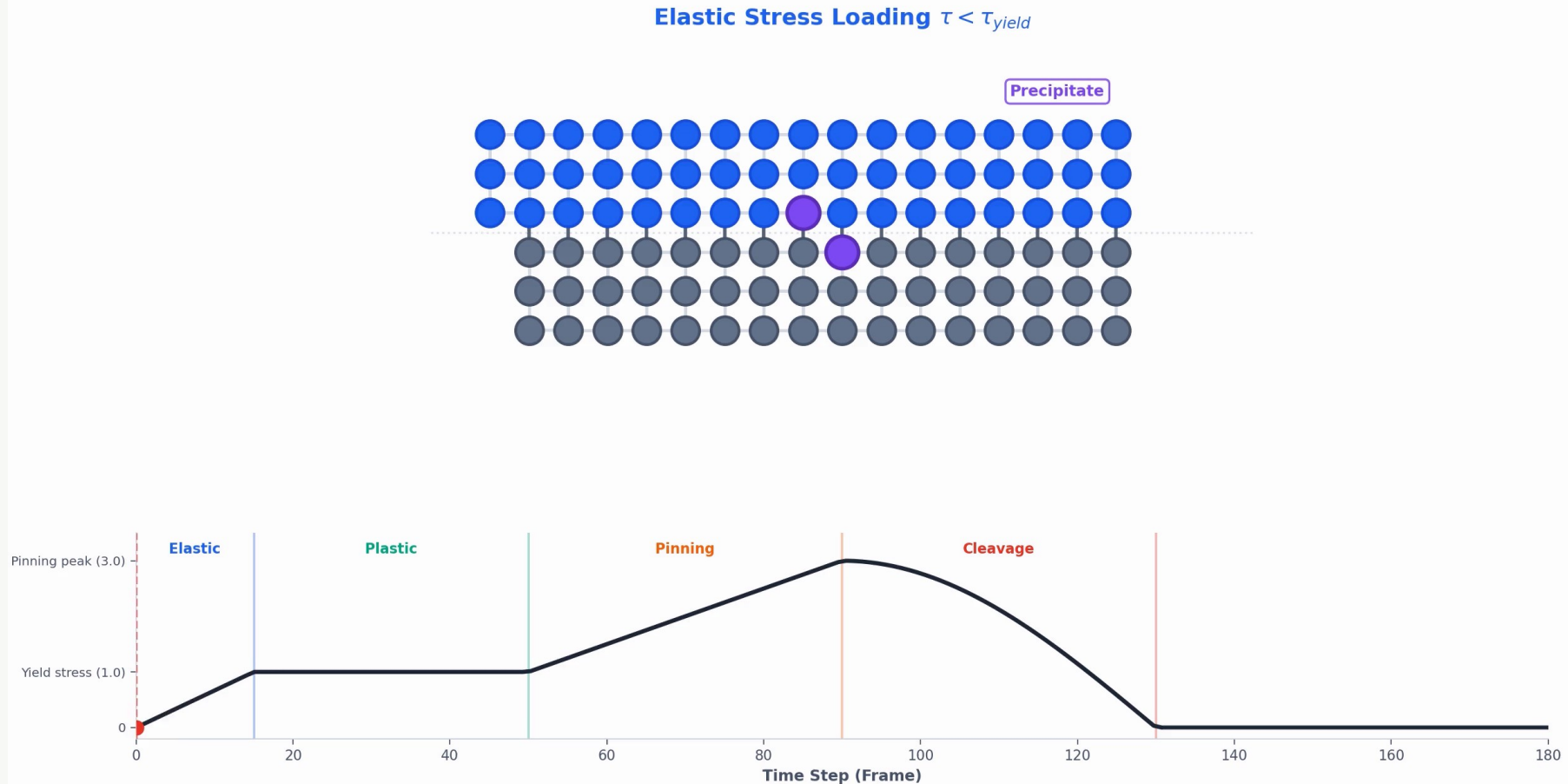


Steel Bends Because Dislocations Move

- Dislocations let steel deform without breaking every bond at once.
- The atoms slide step by step.
- This is the easy route for plastic deformation



Radiation Defects Block That Motion



PAS Enters Here

Motivation

Shortcut

Measurement

Evidence

Takeaway

1. Neutron Bombardment

Creates Vacancies And Dislocations In The Steel Lattice.



2. Vacancy-Type Defects

Small Open-Volume Defects And Clusters Form.



3. Defect Clusters / Trapped Impurities

Local Obstacles Build Up Around The Damaged Lattice.



4. Dislocation Pinning

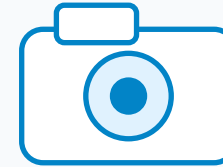
The Easy Route For Plastic Deformation Is Blocked.



5. Brittle Fracture Risk

Stress Concentrates At Cracks Instead Of Being Relaxed.

PAS Enters Here



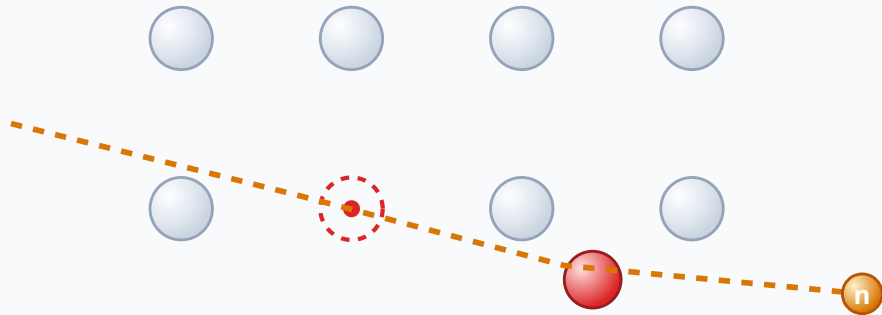
Measures

Positron Lifetime Is Sensitive
To Vacancy-Type Defects.

The Fast Surrogate

Real Neutron Damage

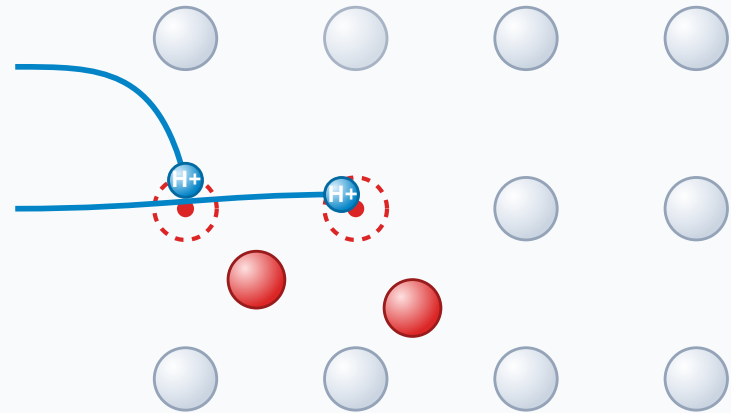
Reactor Environment (Decades)



- Uniform Bulk Damage Across Thick Walls
- Timeframe: **30 To 40 Years**

Simulated Proton Damage

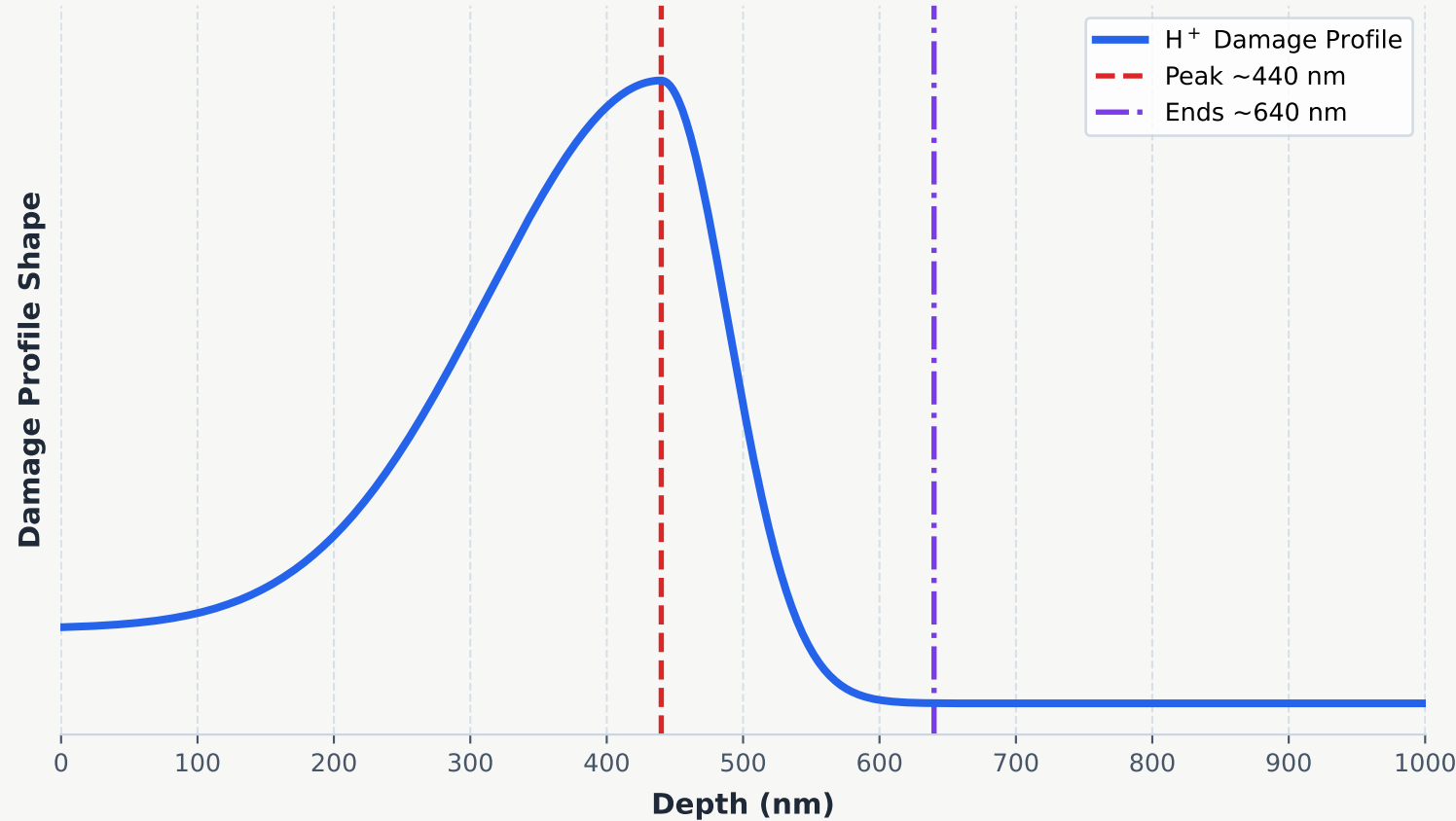
Ion Beam Accelerator (Accelerated)



- Extremely Shallow Damage Zone ($< 1 \mu\text{m}$)
- Timeframe: **Just A Few Hours**

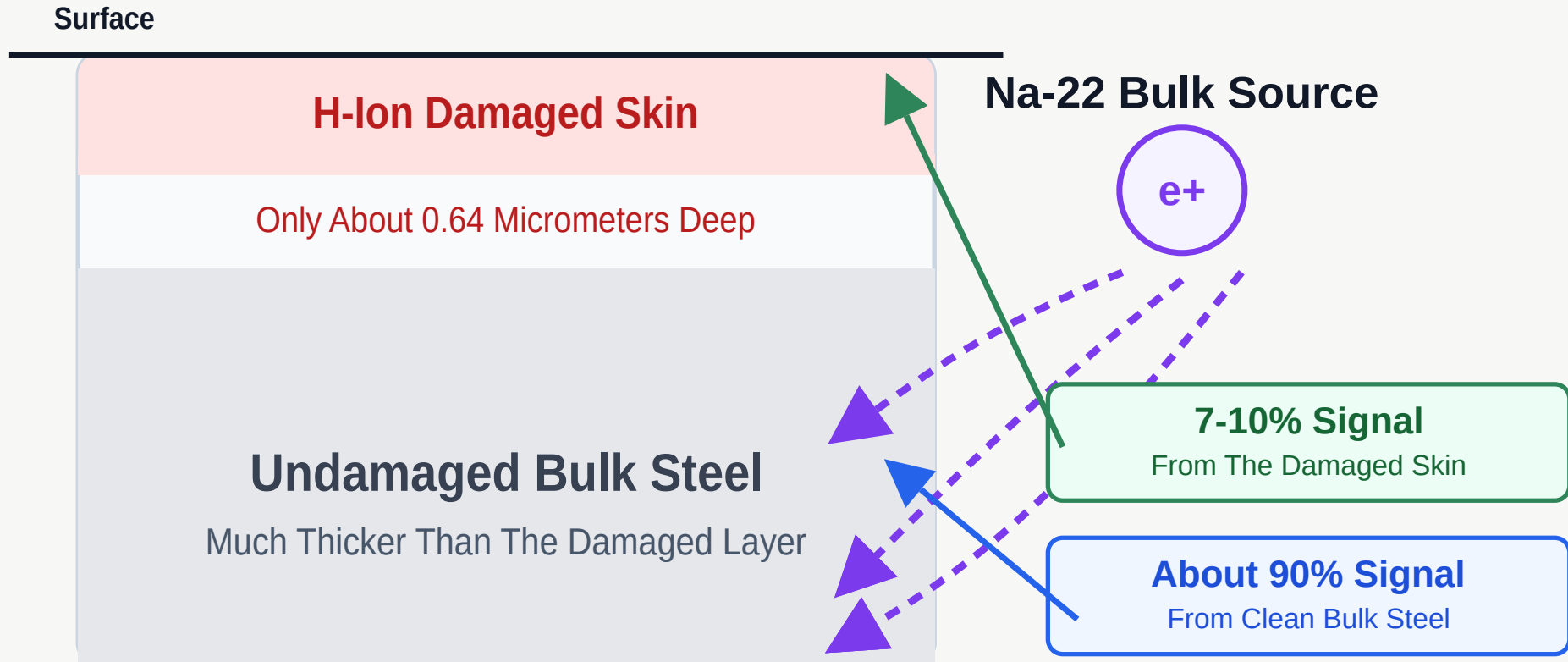
H-Ion Damage Is a Shallow Skin

Where The H^+ Damage Sits



Bulk PALS Dilutes the Signal

The Damage Is Near the Surface. Not good match-up



Pulsed Low Energy Positron System

PLEPS

Pulsed

Controls Lifetime

Beam Arrives In Short Bursts

Low-Energy

Controls Depth

Tunable Energy Sets Stopping Depth

Positron System

Controls Background

Slow-Positron Beamline, Not Na-22

PAS Plot Reading Table

Parameter	Physical Meaning	Interpretation
τ	Average Lifetime	Longer $\rightarrow$ More Open Volume
τ_2	Defect Lifetime	Bigger $\rightarrow$ Larger Defect Size
I_2	Defect Intensity	Higher $\rightarrow$ Higher Defect Density
τ_1, I_1	Bulk Lattice	Defect-Free Lattice

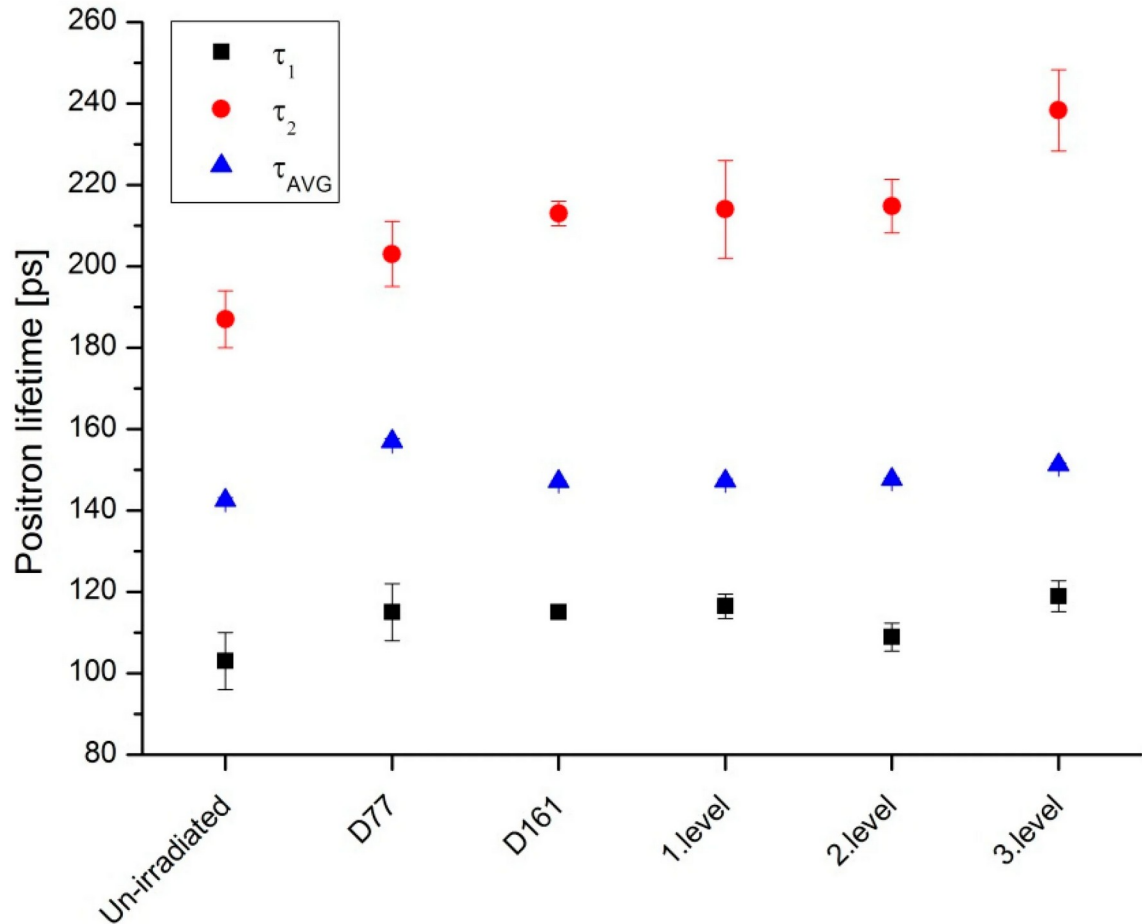
PAS Plot Reading Table

Paper Label	Meaning	Useful Translation
Un-irradiated	Reference Steel	Baseline
D77, D161	Neutron-Irradiated Steel	Real Neutron Damage
1.Level	H-Ion Implantation Level 1	0.5 dpa Local Layer
2.Level	H-Ion Implantation Level 2	4 dpa Local Layer
3.Level	H-Ion Implantation Level 3	15 dpa Local Layer

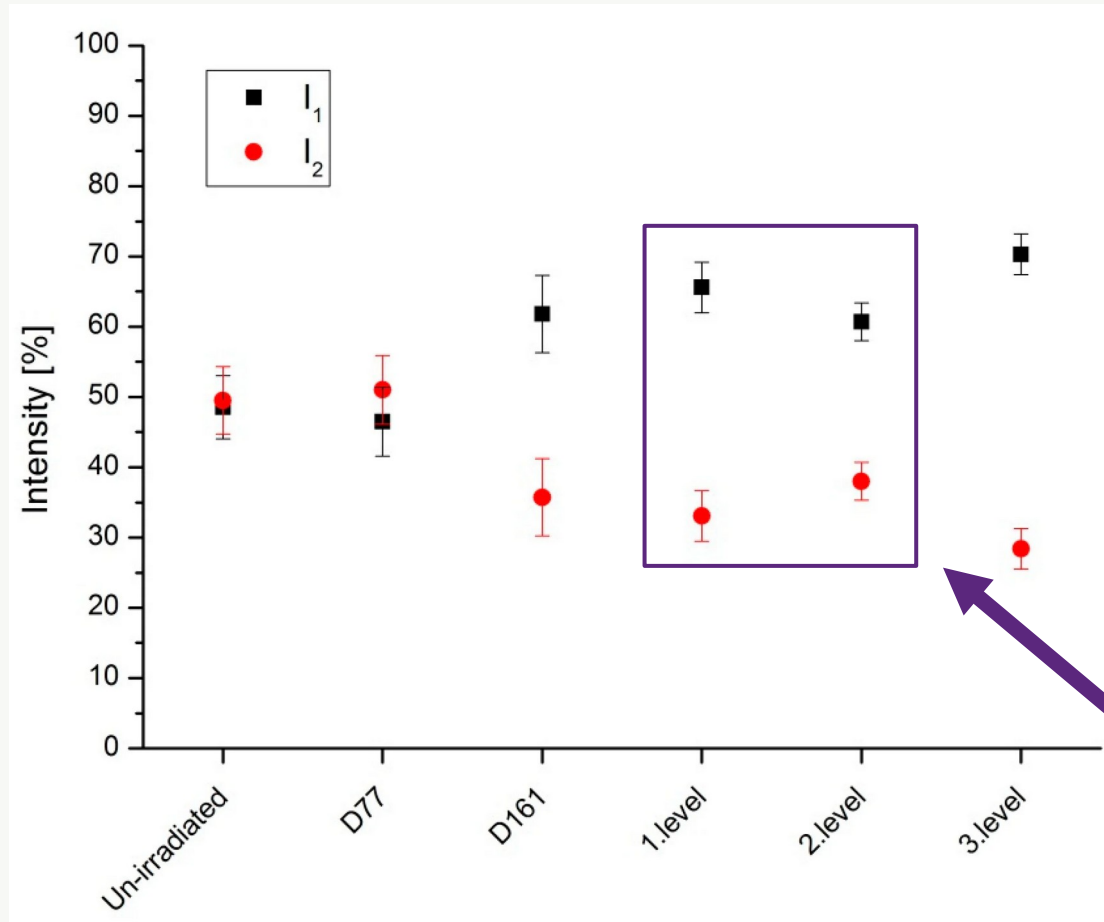
Do the Lifetimes Look Similar?

- Longer τ_2 = more open volume in the defect component
- Real samples: About **203-213 [ps]**
- H-ion Level 1 & 2: Similar size range
- H-ion Level 3: Larger vacancy clusters.

This plot starts the question. It does not finish it.



What About The Defect Fraction?



Stopping and Range of Ions in Matter



Figure 5: Where the Shortcut Stops Being Simple

Key Plot

- Green: Implanted Hydrogen Profile
- Grey/black: Measured PLEPS Lifetime Change
- *What Author Expected:* dpa Peak $\rightarrow$ Vacancy Damage $\rightarrow$ Lifetime peak
- Actual: Lifetime Does Not Follow dpa Alone.

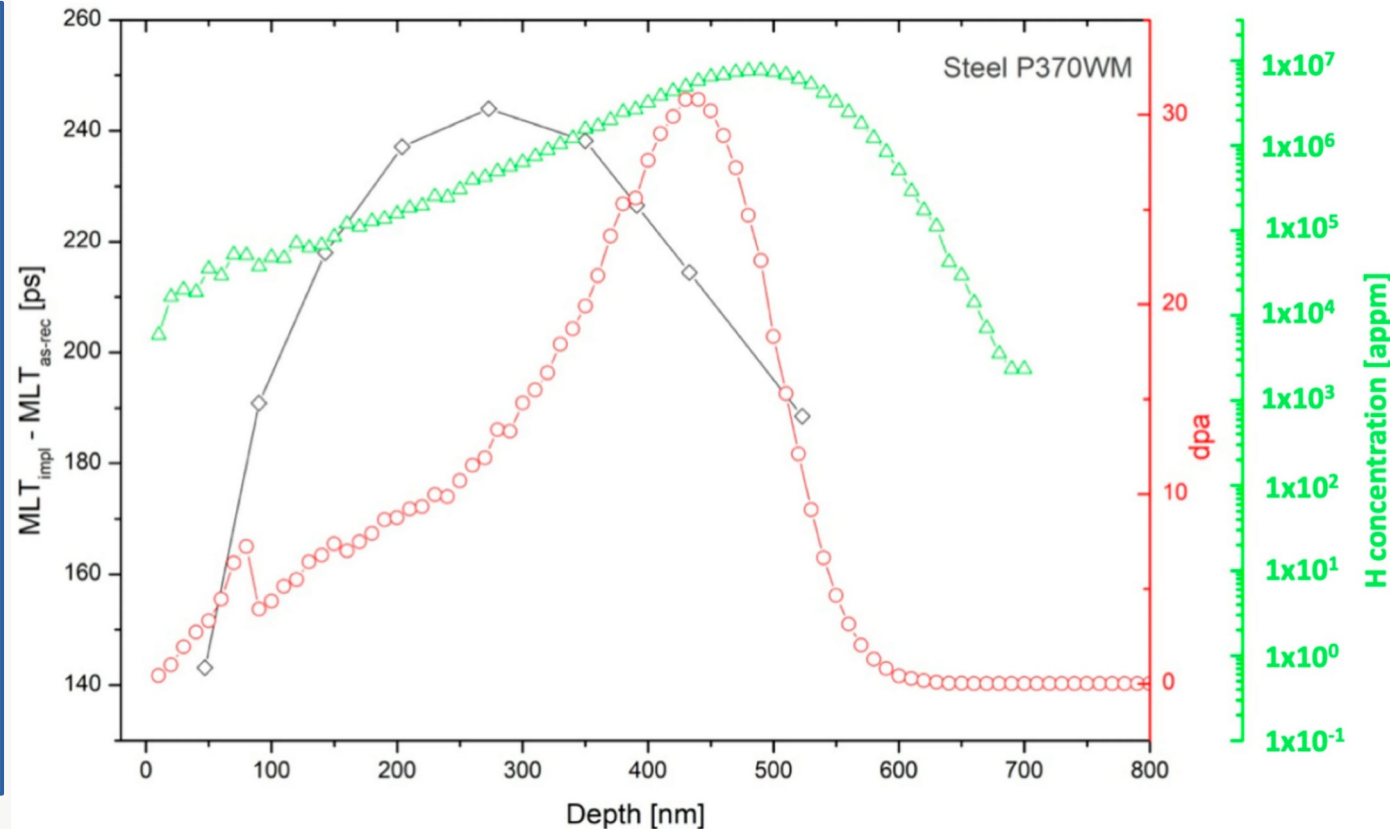
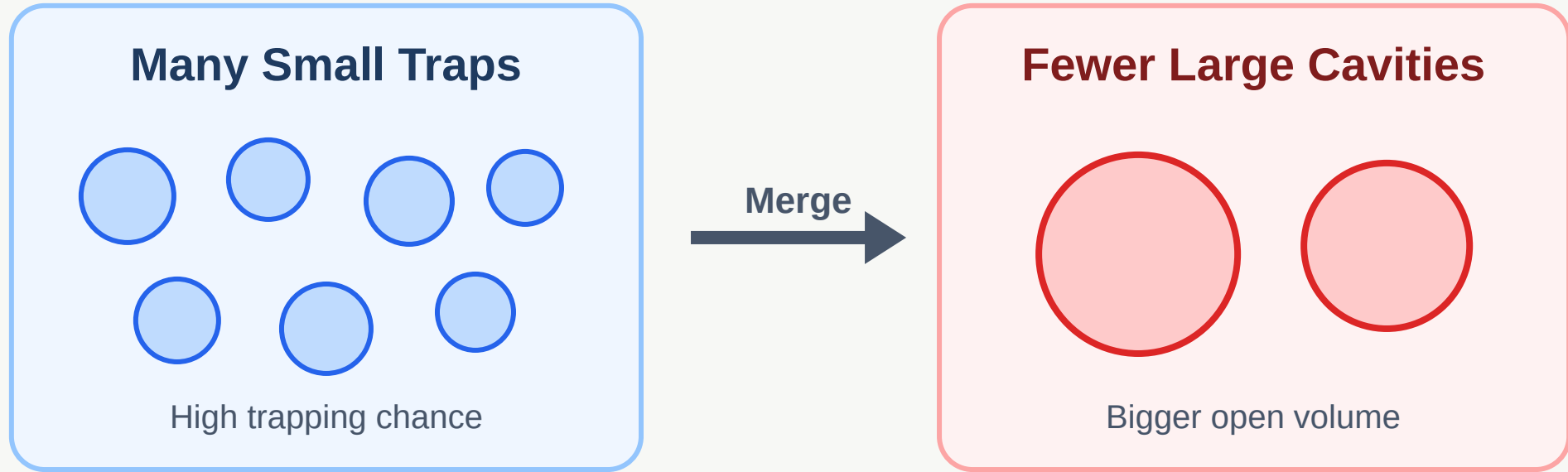


Figure 2 Clue: Fewer Traps, Larger Cavities

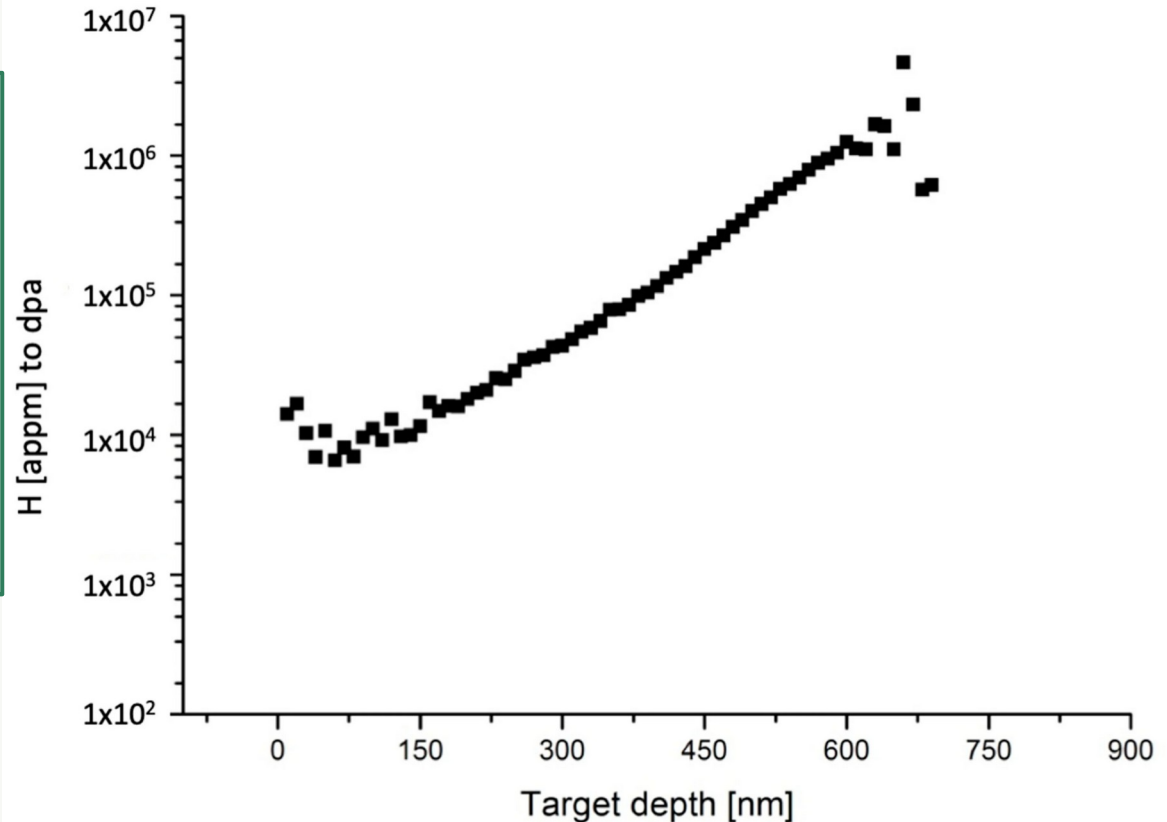


- Fewer traps
- Larger Cavities
- This points to vacancy coalescence

$\uparrow \tau_2 \downarrow I_2$

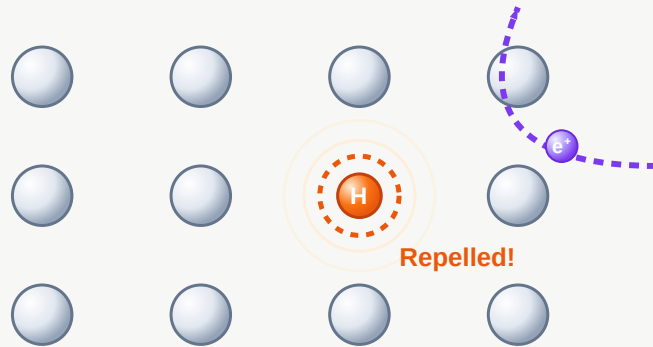
Hydrogen Changes Which Defects Survive

- The H/dpa ratio changes with depth
- Neutrons: damage only
- H ions: damage + injected hydrogen
- H-ion Level 3: Larger vacancy clusters.

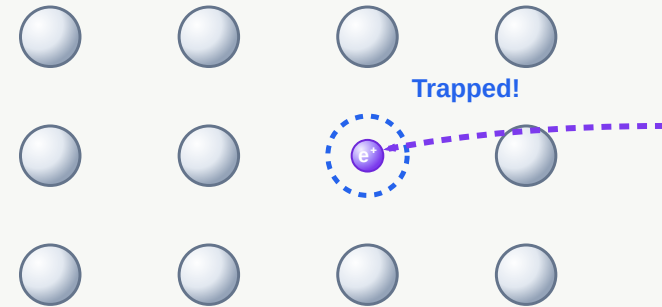


Why Hydrogen Decoration Blocks Positron Trapping

Vacancy Decorated By H



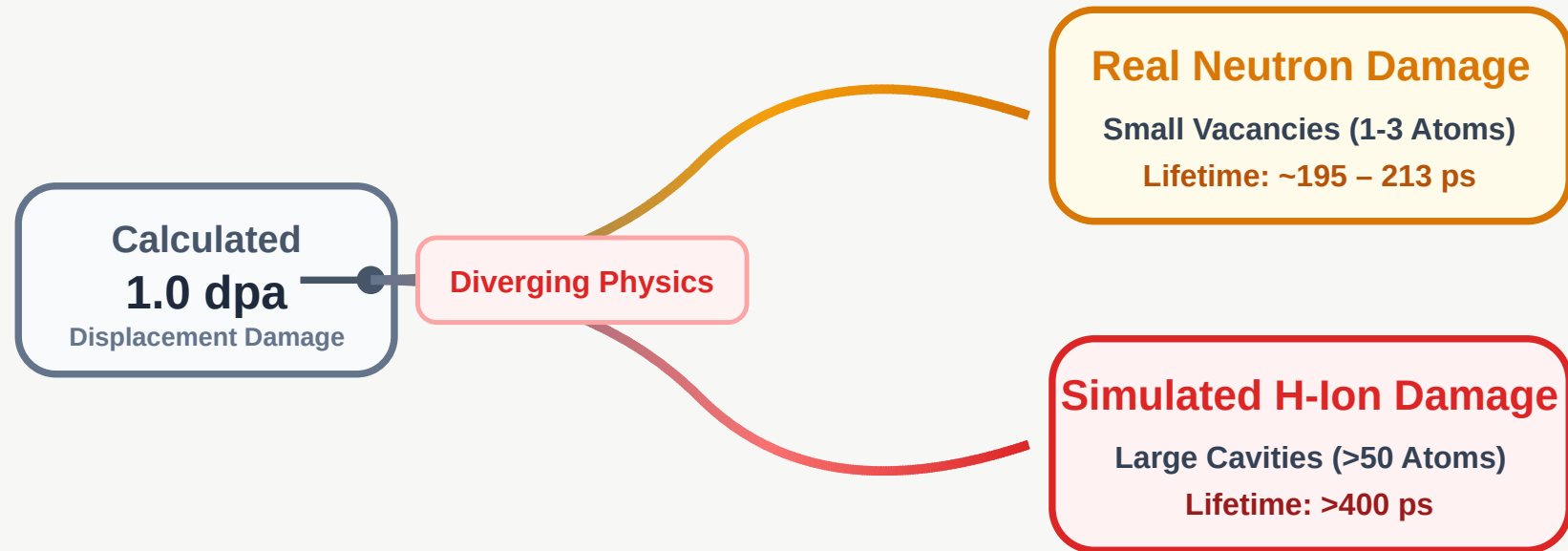
Clean Vacancy



Matching dpa is not enough

The Limits Of Emulation

Why Identical Displacement Damage Calculations Fail To Predict Actual Defect Physics



Takeaway: Fast Ion Implantation is a useful surrogate, but leaves hydrogen behind, altering defect chemistry.

Final Takeaway

Motivation

Background

Mechanism

Evidence

Takeaway

PLEPS makes the shallow ion-damaged layer measurable



H-ion implantation is useful, but not a perfect copy



Hydrogen changes the vacancy structure

Questions?